Variational Quantum Cellular Automata

$E_j(\theta)$ Transition Function Circuit Designs

1 Transition Circuit Code 12

$$E_j(\theta)\,|\psi\rangle = CRX_{i,w}(\theta_2) \cdot CRX_{k,w}(\theta_1) \cdot CX_{j,w} \cdot |\psi\rangle$$

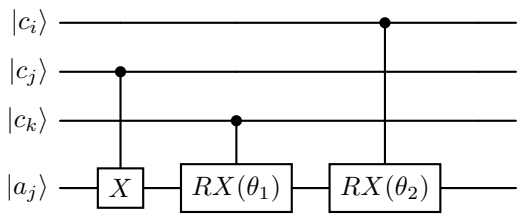


Figure 1: Circuit 21

2 Transition Circuit Code 22

$$E_j(\theta)\,|\psi\rangle = CRY_{i,w}(\theta_2) \cdot CRY_{k,w}(\theta_1) \cdot CX_{j,w} \cdot |\psi\rangle$$

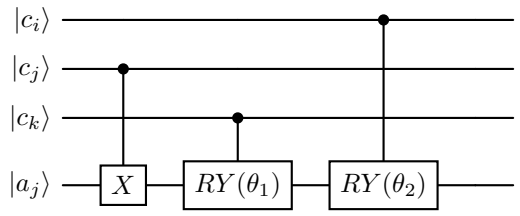


Figure 2: Circuit 22

3 Transition Circuit Code 31

$$E_j(\theta)\,|\psi\rangle = RX_w(\theta_3) \cdot CRX_{i,w}(\theta_2) \cdot CRX_{k,w}(\theta_1) \cdot CX_{j,w} \cdot |\psi\rangle$$

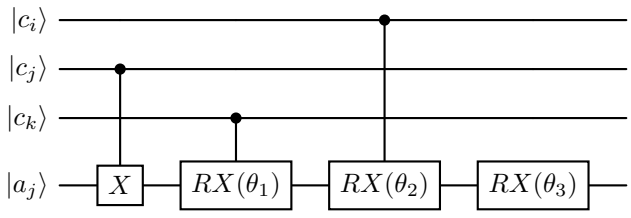


Figure 3: Circuit 31

4 Transition Circuit Code 32

$$E_j(\theta)\,|\psi\rangle = RY_w(\theta_3) \cdot CRY_{i,w}(\theta_2) \cdot CRY_{k,w}(\theta_1) \cdot CX_{j,w} \cdot |\psi\rangle$$

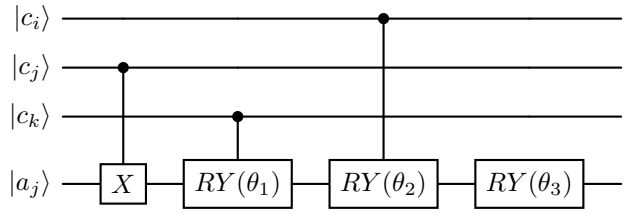


Figure 4: Circuit 32

5 Transition Circuit Code 33

$$E_j(\theta)\,|\psi\rangle = CRX_{i,w}(\theta_3) \cdot CRX_{k,w}(\theta_2) \cdot CRX_{j,w}(\theta_1) \cdot |\psi\rangle$$

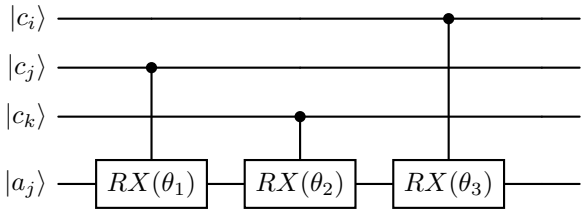


Figure 5: Circuit 33

6 Transition Circuit Code 34

$$E_j(\theta)\,|\psi\rangle = CRY_{i,w}(\theta_3) \cdot CRY_{k,w}(\theta_2) \cdot CRY_{j,w}(\theta_1) \cdot |\psi\rangle$$

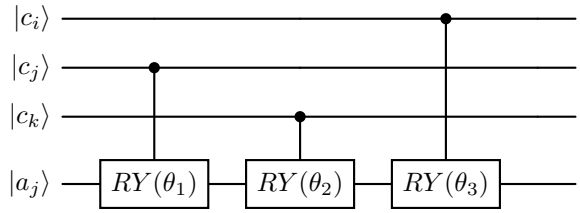


Figure 6: Circuit 34

7 Transition Circuit Code 41

$$E_j(\theta)\,|\psi\rangle = RX_w(\theta_4) \cdot CRX_{i,w}(\theta_3) \cdot CRX_{k,w}(\theta_2) \cdot CRX_{j,w}(\theta_1) \cdot |\psi\rangle$$

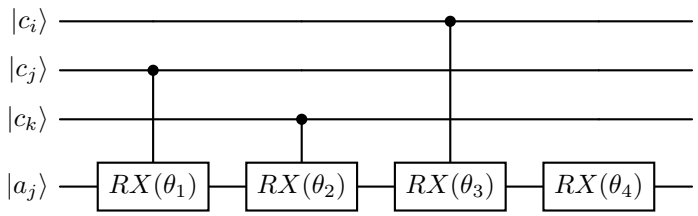


Figure 7: Circuit 41

8 Transition Circuit Code 42

$$E_j(\theta)\,|\psi\rangle = RX_w(\theta_4) \cdot CRX_{i,w}(\theta_3) \cdot CRX_{k,w}(\theta_2) \cdot CRX_{j,w}(\theta_1) \cdot |\psi\rangle$$

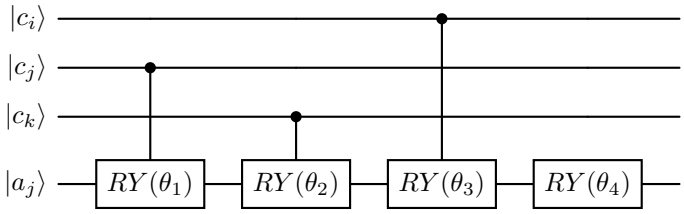


Figure 8: Circuit 42

9 Transition Circuit Code 61

$$E_j(\theta)\,|\psi\rangle = CR_{i,w}(\theta_4, \theta_5, \theta_6) \cdot CR_{k,w}(\theta_1, \theta_2, \theta_3) \cdot CX_{j,w} \cdot |\psi\rangle$$

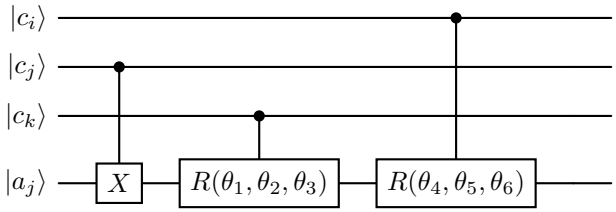


Figure 9: Circuit 61

10 Transition Circuit Code 91

$$E_j(\theta)\,|\psi\rangle = R_w(\theta_7, \theta_8, \theta_9) \cdot CR_{i,w}(\theta_4, \theta_5, \theta_6) \cdot CR_{k,w}(\theta_1, \theta_2, \theta_3) \cdot CX_{j,w} \cdot |\psi\rangle$$

11 Transition Circuit Code 92

$$E_j(\theta)\,|\psi\rangle = CR_{i,w}(\theta_7, \theta_8, \theta_9) \cdot CR_{k,w}(\theta_4, \theta_5, \theta_6) \cdot CR_{j,w}(\theta_1, \theta_2, \theta_3) \cdot |\psi\rangle$$

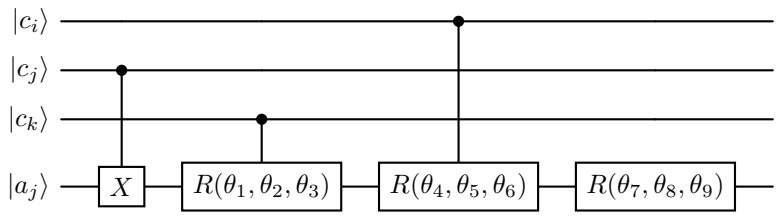


Figure 10: Circuit 91

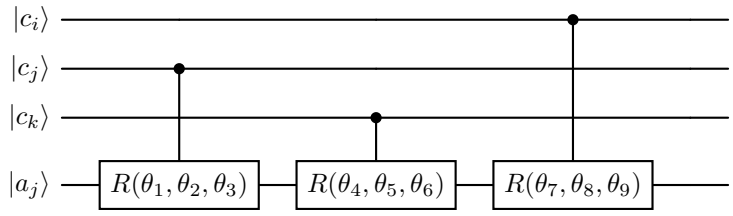


Figure 11: Circuit 92

12 Transition Circuit Code 120

$$E_j(\theta)\,|\psi\rangle = R_w(\theta_{10}, \theta_{11}, \theta_{12}) \cdot CR_{i,w}(\theta_7, \theta_8, \theta_9) \cdot CR_{k,w}(\theta_4, \theta_5, \theta_6) \cdot CR_{j,w}(\theta_1, \theta_2, \theta_3) \cdot |\psi\rangle$$

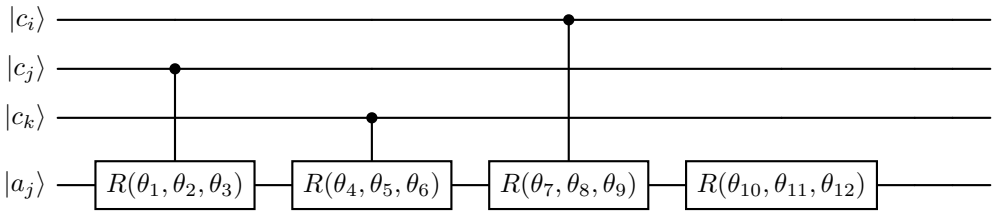


Figure 12: Circuit 120

ID	#Par	Circuit
14	15	$CX_{j,w} \cdot CU_{j,w}(\theta_1, \theta_2, \theta_3) \cdot CU_{k,w}(\theta_4, \theta_5, \theta_6) \cdot U_w(\theta_7, \theta_8, \theta_9) \cdot CU_{k,w}(\theta_{10}, \theta_{11}, \theta_{12}) \cdot CU_{i,w}(\theta_{13}, \theta_{14}, \theta_{15}) \cdot \|\psi\rangle$
15	18	$CU_{j,w}(\theta_1, \theta_2, \theta_3) \cdot CU_{j,w}(\theta_4, \theta_5, \theta_6) \cdot CU_{k,w}(\theta_7, \theta_8, \theta_9) \cdot U_w(\theta_{10}, \theta_{11}, \theta_{12}) \cdot CU_{k,w}(\theta_{13}, \theta_{14}, \theta_{15}) \cdot CU_{i,w}(\theta_{16}, \theta_{17}, \theta_{18}) \cdot \|\psi\rangle$

Table 1: State transition circuit designs